First Exam Review Sheet − Math 102A

- The first exam will be held on **Saturday, March 25, 02:30 − 03:10 pm Electronic Exam**.

Question One: Find the domain of the following functions:

1. $f(x)=\sqrt{x^2+x-2}$. **2.** $f(x)=\dfrac{x}{|x-1|-4}$. **3.** $f(x)=\sqrt{\dfrac{1}{x}-x}$.

Ans. 1. $(-\infty,-2]\cup[1,\infty)$ **2.** $\mathbb{R}-\{-3,5\}$ **3.** $(-\infty,-1]\cup(0,1]$.

Question Two: Find the range of the following functions:

1. $f(x)=1+\sqrt{4-x^2}$. **2.** $f(x)=1-\sqrt{2x-6}$. **3.** $f(x)=\dfrac{2x-1}{x+1}$. **4.** $f(x)=2x-x^2-6$.

Ans. 1. $[1,3]$ **2.** $(-\infty,1]$ **3.** $\mathbb{R}-\{2\}$ **4.** $(-\infty,-5]$.

Question Three: Solve the following inequalities:

1. $1-x\le 2x-3<6-3x$. **2.** $x^2-2x<3$. **3.** $\dfrac{x}{x+1}\ge x$. **4.** $|2x-3|>1$.

Ans. 1. $\left[\frac{4}{3},\frac{9}{5}\right)$ **2.** $(-1,3)$ **3.** $(-\infty,-1)$ **4.** $(-\infty,1)\cup(2,\infty)$.

Question Four: Equation of Lines and Circles:

1. Find the equation of the line passing through the point $(-1,2)$ and parallel to $x-2y=4$.
2. Find the equation of the circle centred at $(-1,2)$ and passing through the point $(2,5)$.
3. Find the equation of the line passing through the center of the circle $x^2+y^2-2y-3=0$ and the point $(1,2)$.

Ans. 1. $y=\frac{1}{2}x+\frac{5}{2}$ **2.** $(x+1)^2+(y-2)^2=18$ **3.** $y=x+1$.

Question Five: Operations on Functions:

1. If $f(x)=\sqrt{4-2x}$ and $g(x)=\sqrt{x}-1$. Find the domain of $\left(\dfrac{f}{g}\right)(x)$.
2. If $f(x)=\dfrac{1}{x+1}$ and $(g\circ f)(x)=\dfrac{1}{x}$. Find $g(x)$.
3. If $f(x)=\dfrac{1}{x-2}$ and $g(x)=\sqrt{x-1}$. Find the domain of $(g\circ f)(x)$ and $(f\circ g)(x)$.

Ans. 1. $[0,1)\cup(1,2]$ **2.** $g(x)=\frac{x}{1-x}$ **3.** $(2,3]$ and $[1,5)\cup(5,\infty)$.

Question Six: Find the graphical solution for the following system of linear inequalities:

$$y>2x,\qquad 2y+x\ge 1,\qquad y<2.$$